

Underwriter Class Cheat Sheet

 `Underwriter(*, name='Rory', databases=None, update=cfg, log2=cfg, debug=False)`^[1]

The Underwriter manages the creation of Aggregate, Severity, Portfolio, and Distortion objects, maintains a knowledge base of named DecL declarations keyed by (kind, name), and provides the interface to the Lark-based DecL parser. The convenience method  `build`^[2] is the primary entry point; `uw(program)` delegates to it. The following tables show all

 methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

name identifying name for this Underwriter instance
databases the load request: *name / path / glob*, or 'default', 'user', 'all'; a bare `Underwriter()` loads nothing
update if true, constructed objects are updated (probabilities computed); defaults to `config`
log2 default $\log_2$ of the number of buckets used to discretize; defaults to `config`
debug if true, print parser debug messages

2. Update

 `build`^[2] (primary entry point: parse a single-output DecL program and return the constructed object),
 `build_many` (parse a multi-output program, construct each, smart-update; returns `list[ParsedProgram]`),
knowledge (the knowledge base as a (kind, name)-indexed `DataFrame`, lazy-loaded),  `load` (resolve a request and read matching .agg files into the knowledge),  `add_entry` (add or overwrite one parsed declaration in the knowledge),  `to_agg` (export selected knowledge entries to a .agg file; re-loads cleanly),  `reload` (reset to as-created state and re-read the configured request), *lexer*, *parser* (the DecL lexer / parser objects)

3. Moments

None

4. Statistical functions

None

5. Validation

 `interpret_file` (parse every DecL program in a .agg / .csv file and return per-line error info; collects *all* errors rather than aborting — validate a file before installing it), `test_suite_file` (path to the bundled test suite .agg, or *None*)^[3]

6. Output dataframes

None

7. Reinsurance

None

8. Visualization

None

9. Risk and pricing

None

10. Approximations

None

11. Meta

 `discover` (search the knowledge base by regex and optional kind; optionally describe, plot, or return the objects — the directory view, replacing `qlist/qshow/show/dir`),  `help` (help on methods and properties matching regex),
 `available_databases` (list .agg files on the search path: `cwd / user / default`),  `resolve_databases` (dry run: preview the files a request *would* load),
 `show_settings` (print every resolved setting and where its value came from),  `reload_settings` (re-read the config file / environment),
 `write_default_config` (write the annotated config template to `user_dir`), `default_dir` (bundled .agg databases), `user_dir` (`~/ .aggregate`), `version` (aggregate version string)

Notes:

[1]: Arguments are **keyword-only** (leading *): use `databases='test_suite'`, not `Underwriter('test_suite')`. `update` and `log2` default to the [build] section of `config.toml`.

[2]: Also exported as `from aggregate import build`. The module-level `build` pre-loads `build.databases` (`test_suite`) whereas a bare `Underwriter()` starts empty. `build` returns one object, `build_many` a list.

[3]: `aggregate/agg/test_suite.agg` (categories A–O) is now exercised by `uv run pytest`, not by an Underwriter method.

`kind` ∈ {sev, agg, port, distortion}; database search path `cwd` → `user_dir` → `default_dir`.

